

$$\hat{a}_t = \rho_a \hat{a}_{t-1} + \varepsilon_t^a, \quad \hat{y}_t^n = \psi y_a^n \hat{a}_t \quad \text{where } \psi y_a^n = \frac{1+\phi}{\bar{\sigma}+\phi+\alpha(1-\bar{\sigma})} > 0$$

$$\Rightarrow r_t = \phi y \hat{y}_t^n = \phi y \psi y_a^n \hat{a}_t \quad \hat{r}_t^n = \bar{\sigma} E_t [\Delta \hat{y}_{t+1}^n] \Rightarrow \hat{r}_t^n = -\bar{\sigma} \psi y_a^n (1-\rho_a) \hat{a}_t$$

$$\Rightarrow \hat{r}_t^n - r_t = -\psi y_a^n [\bar{\sigma} (1-\rho_a) + \phi y] \hat{a}_t \quad \Delta \hat{y}_{t+1}^n = \hat{y}_{t+1}^n - \hat{y}_t^n = \psi y_a^n (\hat{a}_{t+1} - \hat{a}_t)$$

$$E_t \Delta \hat{y}_{t+1}^n = \psi y_a^n (\rho_a - 1) \hat{a}_t$$

$$B_T (\hat{r}_t^n - r_t) = [k] \Omega [-\psi y_a^n [\bar{\sigma} (1-\rho_a) + \phi y]] \hat{a}_t \quad \text{Only this term related to } \phi y, \text{ others not}$$

$$\text{Shock loading: } f(\phi y) = \frac{\bar{\sigma} (1-\rho_a) + \phi y}{\bar{\sigma} + \phi y + \kappa \phi \pi} \Rightarrow f'(\phi y) = \frac{\bar{\sigma} \rho_a + \kappa \phi \pi}{(\bar{\sigma} + \phi y + \kappa \phi \pi)^2} > 0$$

Shock term gets larger if the central bank responds more aggressively to output

$\Rightarrow$ output gap and inflation variance $\uparrow$ welfare $\downarrow$

$\phi \pi \Rightarrow +\infty \Rightarrow f(\phi y) \rightarrow 0 \Rightarrow$ the exogenous driving force hitting the system goes to 0

$\Rightarrow$ Aggressive inflation targeting gets arbitrarily close to the optimum

b) A constant money growth rule Friedman (1960): $\Delta \hat{m}_t = 0 \quad \forall t$

$$\hat{m}_t - \hat{p}_t = \hat{y}_t - \eta \hat{i}_t - \zeta_t, \quad \zeta_t \text{ is an exogenous money demand shock: } \Delta \zeta_t = \beta \zeta_{t-1} + \varepsilon_t^\zeta$$

$$\hat{m}_t = \hat{y}_t + \hat{y}_t^n - \eta \hat{i}_t - \zeta_t \quad \text{Let } \hat{m}_t^+ = \hat{m}_t + \zeta_t \text{ denotes real balances adjusted by exogenous component of money demand}$$

$$\Rightarrow \hat{i}_t = \frac{1}{\eta} (\hat{y}_t + \hat{y}_t^n - \hat{m}_t^+)$$

$$\hat{m}_t^+ - \hat{m}_{t-1}^+ = \hat{m}_t - \hat{m}_{t-1} - \hat{p}_t + \hat{p}_{t-1} + \zeta_t - \zeta_{t-1} \Rightarrow \hat{m}_{t-1}^+ = \hat{m}_t^+ + \hat{\pi}_t - \Delta \zeta_t$$

$$A_{M,0} \begin{bmatrix} \hat{y}_t \\ \hat{\pi}_t \\ \hat{m}_{t-1}^+ \end{bmatrix} = A_{M,1} \begin{bmatrix} E_t \hat{\pi}_{t+1} \\ E_t \hat{\pi}_{t+1} \\ \hat{m}_t^+ \end{bmatrix} + B_M \begin{bmatrix} \hat{r}_t^n \\ \hat{y}_t^n \\ \Delta \zeta_t \end{bmatrix}$$

* The degree of stability of money demand is a key element in determining the desirability of a money growth rule.

c) Rotemberg and Woodford (1999). Second order approximation to household utility around efficient allocation

$$\tilde{y}_t = y_t - y_t^n \quad \text{expand utility around } x = \tilde{y}_t = \tilde{c}_t \quad U = \frac{C^{1-\bar{\sigma}}}{1-\bar{\sigma}} - \frac{N^{1+\varphi}}{1+\varphi}$$

$$y_t = A_t N_t^{1-\alpha} \Rightarrow \tilde{y}_t = (1-\alpha) \tilde{n}_t \Rightarrow \tilde{n}_t = \frac{\tilde{y}_t}{1-\alpha} \quad \text{consumption moves by } x, \text{ labor moves by } \frac{x}{1-\alpha}$$

$$\text{Thus, consumption part: } \frac{(C^N e^x)^{1-\bar{\sigma}} - (C^N)^{1-\bar{\sigma}}}{(1-\bar{\sigma}) U_C^N C^N} \Rightarrow (*) = \frac{e^{(1-\bar{\sigma})x} - 1}{1-\bar{\sigma}} \Rightarrow (*) = x + \frac{1}{2} (1-\bar{\sigma}) x^2$$

$$U_C^N = (C^N)^{-\bar{\sigma}}$$

$$e^{(1-\bar{\sigma})x} \approx 1 + (1-\bar{\sigma})x + \frac{1}{2} (1-\bar{\sigma})^2 x^2$$

Taylor expansion around 0

6 Labor part: Since consumption move by χ , labor move by $\chi/(1-\alpha)$

Expand labor by $\frac{\chi}{1-\alpha}$ gives the normalized labor utility change:

$$- \frac{(N^n e^{\frac{\chi}{1-\alpha}})^{1+\varphi} - (N^n)^{1+\varphi}}{(1+\varphi) U_c^n C^n} \quad (**)$$

Since $-\frac{U_N}{U_C} = MPN = (1-\alpha) \frac{Y^n}{N^n}$
 $U_N = (N^n)^\varphi$ } $\Rightarrow \frac{(N^n)^\varphi}{U_c^n C^n} = (1-\alpha) \frac{C^n}{N^n} \Rightarrow \frac{(N^n)^{1+\varphi}}{U_c^n C^n} = (1-\alpha)$

$$\Rightarrow (**) = - \frac{e^{(1+\varphi)\chi/(1-\alpha)} - 1}{1+\varphi} \frac{(N^n)^{1+\varphi}}{U_c^n C^n} = - (1-\alpha) \frac{e^{(1+\varphi)\chi/(1-\alpha)} - 1}{1+\varphi}$$

$$e^{(1+\varphi)\chi/(1-\alpha)} \approx 1 + \frac{1+\varphi}{1-\alpha} \chi + \frac{1}{2} \left(\frac{1+\varphi}{1-\alpha}\right)^2 \chi^2$$

expand at $\chi=0$

$$(**) = - \frac{1-\alpha}{1+\varphi} \left(\frac{1+\varphi}{1-\alpha} \chi + \frac{1}{2} \left(\frac{1+\varphi}{1-\alpha}\right)^2 \chi^2 \right) = -\chi - \frac{1}{2} \frac{1+\varphi}{1-\alpha} \chi^2$$

6 Combine (consumption part + labor part) = $\chi + \frac{1}{2}(1-\bar{\sigma})\chi^2 - \chi - \frac{1}{2} \frac{1+\varphi}{1-\alpha} \chi^2$
 $= \frac{1}{2} \left(1-\bar{\sigma} - \frac{1+\varphi}{1-\alpha} \right) \chi^2 = -\frac{1}{2} \left(\bar{\sigma} - 1 + \frac{1+\varphi}{1-\alpha} \right) \chi^2 = -\frac{1}{2} \left(\bar{\sigma} + \frac{\alpha+\varphi}{1-\alpha} \right) \tilde{y}^2$
 $(\chi = \tilde{y})$

6 Now derive the inflation part: $P_t(c_i) \neq P_t(\bar{c})$ inflation creates relative-price dispersion

Cross sectional variance of log relative price $\Rightarrow V = \int_0^1 (p(c_i) - p)^2 di = \int_0^1 s(c_i)^2 di$

$$Y(c_i) = \left(\frac{P(c_i)}{P}\right)^{-\varepsilon} Y \Rightarrow \log Y(c_i) = -\varepsilon \log P(c_i) + \varepsilon \log P + \log Y$$

$$\Rightarrow y(c_i) - y = -\varepsilon (p(c_i) - p) \cdot \frac{s(c_i) = p(c_i) - p}{p} = -\varepsilon s(c_i)$$

$$Y(c_i) = A N(c_i)^{1-\alpha} \Rightarrow N(c_i) = \left(\frac{Y(c_i)}{A}\right)^{\frac{1}{1-\alpha}} \Rightarrow N(c_i) = \left(\frac{Y}{A}\right)^{\frac{1}{1-\alpha}} e^{-\frac{\varepsilon}{1-\alpha} s(c_i)}$$

$$N = \int_0^1 N(c_i) di = \left(\frac{Y}{A}\right)^{\frac{1}{1-\alpha}} \int_0^1 e^{-\frac{\varepsilon}{1-\alpha} s(c_i)} di \quad \text{① Without price dispersion } s(c_i) = 0$$

$$\Rightarrow N^{sym} = \left(\frac{Y}{A}\right)^{\frac{1}{1-\alpha}} \dots \text{symmetric labor/price ②}$$

$$\frac{\text{①}}{\text{②}} = \frac{N}{N^{sym}} = \int_0^1 e^{-\frac{\varepsilon}{1-\alpha} s(c_i)} di \quad \text{let } a = \frac{\varepsilon}{1-\alpha} \Rightarrow \frac{N}{N^{sym}} = 1 - a \int_0^1 s(c_i) di + \frac{a^2}{2} \int_0^1 s(c_i)^2 di$$

$$e^{-a s(c_i)} \approx 1 - a s(c_i) + \frac{a^2}{2} s(c_i)^2$$

expand around $s(c_i) = 0$

(***)

$$P = \left[\int_0^1 P(i)^{1-\varepsilon} di \right]^{\frac{1}{1-\varepsilon}} \Rightarrow 1 = \left[\int_0^1 \left(\frac{P(i)}{P} \right)^{1-\varepsilon} di \right]^{\frac{1}{1-\varepsilon}} \Rightarrow 1 = \int_0^1 e^{(1-\varepsilon)s(i)} di \quad (82/52)$$

$$\frac{P(i)}{P} = e^{p(i)-p} = e^{s(i)}$$

$$e^{(1-\varepsilon)s(i)} = 1 + (1-\varepsilon)s(i) + \frac{1}{2}(1-\varepsilon)^2 s(i)^2 \Rightarrow 1 = \int_0^1 [1 + (1-\varepsilon)s(i) + \frac{1}{2}(1-\varepsilon)^2 s(i)^2] di$$

expand at $s(i)=0$

$$\Rightarrow 1 = 1 + (1-\varepsilon) \int_0^1 s(i) di + \frac{1}{2}(1-\varepsilon)^2 \int_0^1 s(i)^2 di$$

$= V$

$$\Rightarrow \int_0^1 s(i) di + \frac{1}{2}(1-\varepsilon)V = 0 \Rightarrow \int_0^1 s(i) di = -\frac{\varepsilon-1}{2}V$$

$$(***) = \frac{N}{N^{sym}} = 1 - \alpha \frac{\varepsilon-1}{2}V + \frac{\alpha^2}{2}V = 1 + \frac{\varepsilon(1-\alpha+\alpha\varepsilon)}{2(1-\alpha)^2}V \quad (***)$$

Price dispersion increases labor required to produce the same aggregate output

$$-\frac{U_N}{U_C} = MPN = (1-\alpha)\frac{Y}{N} \quad \text{We need to know extra labor's utility loss.}$$

$$\frac{MPN \Delta N}{Y} = (1-\alpha) \frac{\Delta N}{N}$$

$$(**) \Rightarrow \frac{N - N^{sym}}{N^{sym}} = \frac{N}{N^{sym}} - 1 = \frac{\varepsilon(1-\alpha+\alpha\varepsilon)}{2(1-\alpha)^2}V$$

loss from price dispersion

$$\Rightarrow \frac{MPN \Delta N}{Y} = \frac{\varepsilon(1-\alpha+\alpha\varepsilon)}{2(1-\alpha)^2}V$$

$$\text{Define } (H) = \frac{1-\alpha}{1-\alpha+\alpha\varepsilon} \Rightarrow \frac{MPN \Delta N}{Y} = \frac{1}{2} \frac{\varepsilon}{(H)} V$$

(same as lecture 8)

Now convert price dispersion to inflation. Since $P_t^{1-\varepsilon} = (1-\theta)P_t^{*1-\varepsilon} + \theta P_{t-1}^{1-\varepsilon}$

$$\Rightarrow p_t = (1-\theta)p_t^* + \theta p_{t-1} \Rightarrow \pi_t = p_t - p_{t-1} = (1-\theta)(p_t^* - p_{t-1}) = (1-\theta)(p_t^* - p_t) + (1-\theta)\pi_t$$

$$\Rightarrow p_t^* - p_t = \frac{\theta}{1-\theta}\pi_t \Rightarrow s_t^* = \frac{\theta}{1-\theta}\pi_t \quad \text{Non-price-resetters have } \begin{cases} s_t(i) = p_t(i) - p_t \\ s_{t-1}(i) = p_{t-1}(i) - p_{t-1} \\ p_t(i) = p_{t-1}(i) \end{cases}$$

price-setter dispersion

$$\Rightarrow s_t(i) - s_{t-1}(i) = -p_t + p_{t-1} = -\pi_t \Rightarrow s_t(i) = s_{t-1}(i) - \pi_t$$

$$V_t = (1-\theta)(s_t^*)^2 + \theta \int_0^1 (s_{t-1}(i) - \pi_t)^2 di = \frac{\theta^2}{1-\theta}\pi_t^2 + \theta V_{t-1} + \theta \pi_t^2 \quad \text{We start at the end}$$

$$\Rightarrow V_t = \theta V_{t-1} + \frac{\theta}{1-\theta}\pi_t^2 \quad \text{Assume } V_{-1} = 0 \Rightarrow \sum_{t=0}^{\infty} \beta^t V_t = \theta \sum_{t=0}^{\infty} \beta^t V_{t-1} + \frac{\theta}{1-\theta} \sum_{t=0}^{\infty} \beta^t \pi_t^2$$

$$\text{Since } V_{-1} = 0 \Rightarrow \sum_{t=0}^{\infty} \beta^t V_{t-1} = \beta \sum_{t=1}^{\infty} \beta^{t-1} V_{t-1} + V_{-1} = \beta \sum_{t=0}^{\infty} \beta^t V_t$$

$$\Rightarrow \sum_{t=0}^{\infty} \beta^t V_t = \beta \theta \sum_{t=0}^{\infty} \beta^t V_t + \frac{\theta}{1-\theta} \sum_{t=0}^{\infty} \beta^t \pi_t^2 \Rightarrow (1-\beta\theta) \sum_{t=0}^{\infty} \beta^t V_t = \frac{\theta}{1-\theta} \sum_{t=0}^{\infty} \beta^t \pi_t^2$$

$$\Rightarrow \sum_{t=0}^{\infty} \beta^t V_t = \frac{\theta}{(1-\theta)(1-\beta\theta)} \sum_{t=0}^{\infty} \beta^t \pi_t^2 \quad (3)$$

$$\frac{MPN_t \Delta N_t}{Y_t} = \frac{1}{2} \frac{\varepsilon}{H} V_t \Rightarrow \text{Total present value of welfare loss} = \frac{1}{2} \frac{\varepsilon}{H} \sum_{t=0}^{\infty} \beta^t V_t \quad (8253)$$

Plug in (3) $L^\pi = \frac{1}{2} \frac{\varepsilon}{H} \frac{\theta}{(1-\theta)(1-\beta\theta)} \sum_{t=0}^{\infty} \beta^t \pi_t^2$

Define $\lambda = \frac{(1-\theta)(1-\beta\theta)}{\theta} H$ same as lecture 8.

$$L^\pi = \frac{1}{2} \frac{\varepsilon}{\lambda} \sum_{t=0}^{\infty} \beta^t \pi_t^2 = \frac{1}{2} \sum_{t=0}^{\infty} \beta^t \frac{\varepsilon}{\lambda} \hat{\pi}_t^2$$

$$L^C + L^N = \frac{1}{2} \sum_{t=0}^{\infty} \beta^t \left(\bar{\sigma} + \frac{\alpha+\psi}{1-\alpha} \right) \tilde{y}_t^2$$

welfare loss from
consumption + labor

$$\text{Thus } L = L^C + L^N + L^\pi = \frac{1}{2} \sum_{t=0}^{\infty} \beta^t \left[\left(\bar{\sigma} + \frac{\alpha+\psi}{1-\alpha} \right) \tilde{y}_t^2 + \frac{\varepsilon}{\lambda} \hat{\pi}_t^2 \right]$$

Why $\theta \int_0^1 (S_{t+1}(i) - \pi_t)^2 di = \theta V_{t+1} + \theta \pi_t^2$?

$$\int_0^1 (S_{t+1}(i) - \pi_t)^2 di = \int_0^1 S_{t+1}(i)^2 di + \int_0^1 \pi_t^2 di - 2\pi_t \int_0^1 S_{t+1}(i) di$$

Since we have $\int_0^1 S_{t+1}(i) di = \frac{\varepsilon-1}{2} V_{t+1}$

$$\Rightarrow \int_0^1 (S_{t+1}(i) - \pi_t)^2 di = V_{t+1} + \pi_t^2 - (\varepsilon-1) V_{t+1} \pi_t$$

π_t is first-order, V_{t+1} is second-order $\Rightarrow \pi_t V_{t+1}$ is third-order.

In a second-order welfare approximation, we drop third-order terms

$$\Rightarrow \theta \int_0^1 (S_{t+1}(i) - \pi_t)^2 di = \theta V_{t+1} + \theta \pi_t^2$$

$$\text{From } L = \frac{1}{2} \sum_{t=0}^{\infty} \beta^t \left[\left(\bar{\sigma} + \frac{\alpha+\psi}{1-\alpha} \right) \tilde{y}_t^2 + \frac{\varepsilon}{\lambda} \hat{\pi}_t^2 \right]$$

Where $\lambda = \frac{(1-\theta)(1-\beta\theta)}{\theta} H$ $H = \frac{1-\alpha}{1-\alpha+\alpha\varepsilon}$

λ is the slope of NKPC with respect to real marginal cost

Let $\kappa = \lambda \left(\bar{\sigma} + \frac{\psi+\alpha}{1-\alpha} \right)$ -- slope of NKPC with respect to $\tilde{y}_t$

$$L = \frac{1}{2} \sum_{t=0}^{\infty} \beta^t \left[\frac{\kappa}{\lambda} \tilde{y}_t^2 + \frac{\varepsilon}{\lambda} \hat{\pi}_t^2 \right]$$

$$L = \frac{1}{2} \frac{\varepsilon}{\lambda} \sum_{t=0}^{\infty} \beta^t \left[\frac{\kappa}{\varepsilon} \tilde{y}_t^2 + \hat{\pi}_t^2 \right]$$

We let $\alpha_x = \frac{\lambda}{\epsilon}$

$$L = \frac{1}{2} \frac{\epsilon}{\lambda} \sum_{t=0}^{\infty} \beta^t [\hat{\pi}_t^2 + \alpha_x \tilde{y}_t^2]$$

When we have inefficiency at supply side: $\tilde{y}_t = \hat{y}_t - \hat{y}_t^n \neq \hat{y}_t - \hat{y}_t^e = \hat{x}_t$

The welfare loss is coming from the gap between realized current output $\hat{y}_t$ (deviation) and the efficient output $\hat{y}_t^e$ (deviation), thus

$$L = \frac{1}{2} \frac{\epsilon}{\lambda} \sum_{t=0}^{\infty} \beta^t [\hat{\pi}_t^2 + \alpha_x \hat{x}_t^2]$$

Now scale the welfare loss by $2 \frac{\lambda}{\epsilon}$

$$L' = 2 \frac{\lambda}{\epsilon} L = \sum_{t=0}^{\infty} \beta^t [\hat{\pi}_t^2 + \alpha_x \hat{x}_t^2]$$

since $2 \cdot \frac{\lambda}{\epsilon}$ is constant, the scaling does not affect the optimal decision $(\hat{\pi}_t^*, \hat{x}_t^*)$ that minimizes the loss L' .